



2019 Semester 2 Report - Year 12

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
English (Advanced) - HSC	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Design and Technology - HSC	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Engineering Studies - HSC	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Mathematics Extension 1 - HSC	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
Physics - HSC	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			



2019 Semester 2 Report - Year 12

English (Advanced) - HSC

Name: Luke NELSON
Class: 12ENAG2
Teacher: Lisa-Maree Giuffrida

Comment on Progress

Luke is a committed and engaged student, who has worked to the best of his ability and made good progress over the semester. He is to be commended on his mature work ethic, as well as his continual efforts to produce drafts which have helped his to improve his conceptual understanding of texts. Luke's sustained approach to all modules underpins his effective preparation for the Higher School Certificate and I wish him success.

Learning Outcomes

Outcomes	O	H	S	B	L
Critically analyses and uses language forms, features and structures of texts justifying appropriateness for specific purposes, audiences and contexts and evaluates their effects on meaning.		✓			
Strategically adapts and applies knowledge, skills and understanding of language concepts and literary devices in new and different contexts.	✓				
Thinks imaginatively, creatively, interpretively, critically and discerningly to respond to, evaluate and compose texts that synthesise complex information, ideas and arguments.		✓			
Investigates and evaluates the relationships between texts.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2019 Semester 2 Report - Year 12

Design and Technology - HSC

Name: Luke NELSON

Class: 12DTA7

Teacher: Roger Forsyth

Comment on Progress

Luke worked diligently throughout the year to develop an innovative high quality compact skateboard that meets the specific needs of his client. His ability to problem solve and work simultaneously on multiple tasks was commendable. Luke's design folio project documentation clearly communicated his development process and ongoing evaluation in detail. His proficient project management enabled time to check the quality of responses and ensure all criteria were addressed. With ongoing refinement of his exam technique, enhancing his question analysis and composing appropriate responses by completing past HSC papers, Luke can continue to achieve at the highest level in this course.

Learning Outcomes

Outcomes	O	H	S	B	L
Justifies technological activities undertaken, critically assesses the emergence and impact of new technologies, and the factors affecting their development.	✓				
Selects and uses appropriate research methods and communication techniques and manages the development of a quality major design project.	✓				
Evaluates the processes undertaken and the impacts of a major design project that addresses need or opportunity.	✓				
Analyses the factors that influence innovation and the success of innovation, including the use of creative and innovative approaches in designing and producing.	✓				
Explains the influence of trends in society on design and production, and evaluates the impact of design and innovation on society and the environment.	✓				
Critically analyses the factors affecting design and the development and success of design projects, including the practices and processes of designers and producers in relation to the major design project.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2019 Semester 2 Report - Year 12

Engineering Studies - HSC

Name: Luke NELSON

Class: 12ENGSTA1

Teacher: Joanne Drew

Comment on Progress

Luke is a highly motivated and perceptive student who has achieved outstanding results. His flair for design and construction was demonstrated in the development of a hydraulic lifting device and he submitted an excellent report applying many course principles in this practical assessment task. Luke's HSC Trial Examination mark was impressive and reflected his thorough preparation and insightful knowledge of the HSC engineering modules. In order to further improve, he is encouraged to systematically analyse all questions and practise as many drawing questions as possible. Luke is an intelligent, collaborative student and he is wished well for his future endeavours.

Learning Outcomes

Outcomes	O	H	S	B	L
Justifies the selection of materials and components in engineering.	✓				
Analyses engineering applications in specific fields.	✓				
Demonstrates proficiency in the use of mathematical, scientific and graphical methods of analysis.	✓				
Demonstrates an understanding of engineering drawing and drawing standards.		✓			
Demonstrates skill in research and problem solving related to engineering.	✓				
Uses appropriate written, oral and presentation skills in the preparation of detailed engineering reports.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2019 Semester 2 Report - Year 12

Mathematics Extension 1 - HSC

Name: Luke NELSON

Class: 12MAX1D8

Teacher: Marjan Novotni

Comment on Progress

Luke has worked diligently this semester in Mathematics Extension 1 to maintain a consistent level of performance. He is a mature and motivated student, who listens well to explanations and contributes to the development section of lessons. Luke conscientiously completes set tasks and seeks assistance with any difficulties. His work standard is excellent and, as a consequence, his results are very impressive and thoroughly deserved.

Learning Outcomes

Outcomes	O	H	S	B	L
Uses a variety of strategies to investigate mathematical models of situations involving projectiles, velocity and acceleration as functions of displacement and exponential growth and decay.	✓				
Uses the relationship between functions, inverse functions and their derivatives.	✓				
Determines integrals by reduction to a standard form through a given substitution.	✓				
Uses a variety of strategies to investigate mathematical models of situations involving binomial probability and binomial theorem.	✓				
Uses inductive reasoning in the construction of proofs for problems involving circle geometry, parametrics, inequalities and mathematical induction.	✓				
Applies the chain rule to problems involving rates of change.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2019 Semester 2 Report - Year 12

Physics - HSC

Name: Luke NELSON
Class: 12PHYSB4
Teacher: Richard Crooks

Comment on Progress

Luke is a very capable and motivated student whose determination to succeed is reflected in the outstanding results that he attained this semester. He works diligently to extend his knowledge and understanding of key concepts in Physics and participates effectively in both independent tasks and team activities. He carefully analyses and compiles extensive and detailed notes to communicate information succinctly and logically using appropriate scientific formats. Luke always strives to achieve his personal best and is to be congratulated on these excellent grades.

Learning Outcomes

Outcomes	O	H	S	B	L
Processes, analyses and evaluates qualitative and quantitative data and information, from both primary and secondary sources, using a range of appropriate media.	✓				
Solves scientific problems using primary and secondary data, critical thinking skills and scientific processes.	✓				
Communicates scientific understanding using suitable language and terminology for a specific audience or purpose.		✓			
Describes, explains and analyses qualitatively and quantitatively, circular, gravitational and projectile motion.	✓				
Explains and analyses qualitatively and quantitatively electric and magnetic interactions due to charged particles and currents.	✓				
Describes, explains and analyses evidence for the properties of light, and evaluates the implications of this for modern theories of physics including astronomical events.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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25 September 2019

INTERIM REFERENCE

To whom it may concern

Luke Nelson has been a student of Northern Beaches Secondary College, Manly Campus, an academically selective high school, since 2014. Luke is currently a candidate for the NSW Higher School Certificate presenting for English Advanced, Design and Technology, Engineering Studies, Mathematics Extension 1 and Physics.

Throughout Years 10, 11 and 12, Luke has achieved in the areas of the school ethos of *Academic Excellence, Personal Best and Giving Back to the Community* through the following:

- Peer Support Leader 2017
- Technology Leader 2017
- Student Wellbeing Advocate 2018
- Member of the Student Representative Council 2019
- Attended the World Vision Youth Conference 2019
- Chess Captain 2019

These leadership roles have been responsibly fulfilled providing a positive role model to other students.

Luke has been a committed and dedicated student who has achieved school awards in the area of academic studies including:

- Principal's Award 2017 and 2019
- 1st place in Engineering Studies, 3rd place in Chemistry, Academic Excellence in Physics 2018

Luke has also achieved special school awards including:

- Honours Merit Award 2017

These achievements are especially meritorious given the academically selective nature of the school.

In addition to a rigorous academic schedule Luke has demonstrated the ability to successfully balance academic studies with extra-curricular activities through involvement in the following:

- Member of the school's band program 2017 - Big Band
- Duke of Edinburgh Silver Program 2017
- Chess Club 2017-2019

In fulfilling the school ethos of *Giving Back to the Community*, Luke has participated in:

- Gold Volunteering Certificate (60+ hours) 2017
- Endure for a Cure 2017 and 2018
- Wraps for Love 2017

Luke is mature, focused and intelligent young man who has displayed commendable resilience in achieving his goals. He is well respected by his peers and teachers for his calm manner and concern for others. In his involvement in many of the school's extra-curricular activities, he has shown strong commitment to the school community and has become an admirable role model for all students. His admirable personal values, strong sense of ethics and good humour, ensure Luke's success in his future endeavours. He is highly recommended.

Yours sincerely

A handwritten signature in black ink, appearing to read 'Cath Whalan', with a stylized, flowing script.

Cath Whalan
Principal

Luke Nelson

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Education

Primary

Normanhurst Public School

2007-2012

Mona Vale Public School

2012-2013

Secondary

NBSC Manly Selective Campus

2013-Present

Awards

Principal Awards in all years excluding Year 11

2013-2019

Honours Merit Award

2017

HSC Mark of 95 in Advanced Mathematics (Accelerated Course)

2018

1st in Engineering Studies

2018

3rd in Chemistry

2018

Academic Excellence in Physics

2018

Experience

Charity Work

- Participated in the Endure for a Cure charity ride two years in a row, raising over \$1000 collectively for cancer research 2017-2018
- I founded a school-based fundraiser which raised awareness for Autism in schools and donated over \$800 to the ASPECT charity organisation. 2019
- I am currently part of the team organising my school's involvement in the 40 Hour Famine Backpack Challenge. 2019

Community Involvement and Volunteering Work

- I volunteered weekly at the local library to help with the organisation and day to day tasks. 2017
 - I personally designed and programmed an online ordering website for a local website, boosting their sales and market presence (www.musclemunchies.com.au). 2018
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Leadership

- I was a Volunteer Peer Support Leader which provided essential support and information for students transitioning into high school, coaching them on vital study skills and the importance of balance of schooling and personal lives. 2017
- I led and supervised a group of year 4 students in the creation of a multimedia presentation to raise community awareness of the threats facing penguins. 2017
- I was part of a team which designed a technology-based event, teaching younger years for a day through unique activities in virtual and augmented reality. 2017
- I was part of a team of mathematicians who coached high-achieving year 6 students in higher order thinking and problem solving. 2018
- Attended the World Vision Leadership Conference in which I heard talks from prominent civil rights spokespeople. 2019
- Part of the Manly Selective Executive Student Representative Team which directs and supervises the student body in the organisation and production of community and school events, like Pinestock – a yearly celebration which raised over \$2000. 2019
- Chess Captain and lead student organiser of all the school's chess teams. 2019

Work Experience

- Worked at Domino's as a manager for two years, leading the group of workers independently. 2016-2018
- I went to CSIRO and spent a week helping and learning about astronomy and the software requirements under the wing of one of the lead programmers of the ASKAP telescopes. 2017
- I tutor students from Kindergarten to Year 12 in Mathematics both privately and in the organisation NumberWorks'n'Words. 2018-2019

Activities and Interests

Woodworking and Design: Avid designer and creator of products both in school and at home, such as my Design and Technology Major Work of a Compact Skateboard.

Sports: Regular surfer and cyclist; participated in the Endure for a Cure charity ride two years in a row, raising over \$1000 collectively and winning the male student group both times.

Software: Adept at programming in HTML, CSS, Javascript, C & C++; independently created a website for a local business (www.musclemunchies.com.au) and am currently producing a booking service for another.
